

(Please Write your roll No. Immediately)

Roll No.:

First Term Examination, February-2019

B.Tech. (VI sem) CSE/IT

Paper Code : ETCS-302

Subject: Compiler Design

Time : 1.5 hours

Max. Marks:30

Note: Attempt Q. No. 1 which is compulsory and any two other questions from remaining questions. Each question carries 10 marks.

Q.1 a) How dependency graph effects the evaluation order in syntax directed translation?

b) "Lexical Analysis is the most time consuming phase in compilation" Justify this statement.

c) Create NDFA for the regular expression $(a+b)^*abb+bba$.

d) Differentiate between L-Attributed definition and S-Attributed definition.

e) Remove left recursion from the following grammar: $S \rightarrow AaAb | b$ $A \rightarrow Ab | a | ab$ [2*5=10]

Q.2. Prove that following grammar is CLR(1) but not LALR(1)

$S \rightarrow Aa | bAc | Bc | bBa$

$A \rightarrow d$

$B \rightarrow d$

[10]

Q.3.

(a) Calculate first and follow for all non-terminals for the following grammar.

$E \rightarrow E+T | T$

$T \rightarrow TF | F$

$F \rightarrow F* | a | b$

[6]

(b) Create a predictive parsing table for the grammar shown in part a.

[4]

Q.4.

(a) How lex(flex) tool can be used to create lexical analyzer? Explain with the help of an example.

[5]

(b) Define Synthesized and inherited attributes. Identify Synthesized and inherited attributes from the grammar given below:

$D \rightarrow TL$ $L.in = T.type; L.x = T.y$

$L \rightarrow L_1 id$ $L_1.in = L.in; L.x = L_1.x$

$T \rightarrow id$

Justify your answer.

[5]

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Note: Attempt Q. No. 1 which is compulsory and any two other questions from remaining questions. Each question carries 10 marks.

Q.1 a) List the various phases of compilation process.

b) Generate a pattern matcher which can recognize following two patterns: 01^* and 1^*0 .

c) Improve the grammar by removing left recursion. $S \rightarrow AaB \mid b \mid A \rightarrow Aab \mid Aba \mid a \mid B \rightarrow a$.

d) Differentiate between Synthesized attributes and inherited attributes.

e) Define the process used by lex tool for generating lexical analyzer.

[2*5=10]

Q.2. Consider the grammar given below $\epsilon = \text{NULL}$

$$X \rightarrow 0Y1 \mid 1Z2$$

$$Y \rightarrow 2 \mid \epsilon$$

$$Z \rightarrow 1Y2 \mid 1 \mid \epsilon$$

(a) Calculate first and follow for the non-terminals.

[5]

(b) Create a predictive parsing table for the above grammar.

Also Show that this grammar is not LL(1).

[5]

Q.3. Consider the grammar

$$S \rightarrow L=R \mid R$$

$$L \rightarrow *R \mid id$$

$$R \rightarrow L$$

Design LALR parser for this grammar.

[10]

Q.4.

(a) Describe the functioning of YACC with the help of an example.

[5]

(b) Define following terms:

(i) Topological Sort (ii) Dependency Graph (iii) Attribute (iv) Circular SDT

(v) Evaluation order

[5]

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**FIRST TERM EXAMINATION
SIXTH SEMESTER [B.TECH]-FEBRUARY 2017**

Paper Code: ETCS-302

Subject: Compiler Design

Time: 1.5 Hr

Maximum Marks: 30

Note: Attempt three questions. Question No. 1 is compulsory. Each question carries 10 marks.

- Q-1(a) Mention the role of Lexical Analyzer in compiler design. (2)
(b) Explain Chomsky Classification of Languages. (2)
(c) What are the three methods to construct LR parsing tables. Which method is the most powerful? Justify. (2)
(d) An ambiguous grammar can never be LR. Justify the statement with an example. (2)
(e) Design the NFA for the regular expression $a(a+b)^*bb$ (2)

- Q-2 (a) Let G be CFG: (5)

$S \rightarrow bB \mid aA$

$A \rightarrow b \mid bS \mid aAA$

$B \rightarrow a \mid aS \mid bBB$

For the string bbaababa find

- i) leftmost derivation
- ii) rightmost derivation and
- iii) parse tree

- (b) Construct predictive parsing table for the following grammar(S is start symbol):

$S \rightarrow iEtSS_1 \mid a$

$S_1 \rightarrow eS \mid \epsilon$

$E \rightarrow b$

- Q-3 a) Construct SLR parsing table for the following grammar: (5)

$S \rightarrow xAy \mid xBy \mid xAz$

$A \rightarrow aS \mid q$

$B \rightarrow q$

- b) Discuss ambiguity of a CFG. How ambiguity can be removed from the grammar? State suitable example. (5)

- Q4. Write short notes on following :- (2.5 each)

- (a) Bootstrapping and Cross Compiler
- (b) Difference between compiler and interpreter
- (c) Handle and Handle Pruning
- (d) LEX tool

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First-Term Examination

Sixth Semester (B.Tech)

Paper Code: ETCS-302

Time: 1 Hr 30 Mins

Note: Q1 is compulsory and attempt any two questions out of remaining three.

Subject: Compiler Design

Max. Marks: 30

Q1 Attempt any five parts.

(5*2=10)

- Discuss merits and demerits of single pass compiler and multipass compiler.
- What do you mean by bootstrapping in compilation?
- What is the need for separating the parser from scanner?
- Find leftmost derivation, rightmost derivation and derivation tree of a string 'aacbbcc' from the following grammar productions $\{S \rightarrow aAcB | Bds, B \rightarrow aAcA | cAB | b, A \rightarrow aB | aBc | a\}$.
- Explain handle pruning with an example.
- What are the responsibilities of Loader, Linker and Assembler in the compiler environment?
- Write steps involved in compilation of following sentence:
 $pes = interest + rate * 50$

Q2 Attempt both parts

(2*5=10)

- Remove left recursion from the following grammar:

$S \rightarrow Aa | b$

$A \rightarrow Ac | Sd | e$

- Prove whether following grammar is LL(1)

$S \rightarrow XS | dS | e$

$X \rightarrow Y | Zb | aY$

$Y \rightarrow cZ$

$Z \rightarrow e$

$S \rightarrow Aa | b$

$A \rightarrow bdA' | eA'$

$A' \rightarrow AcA' | AadaA' | \epsilon$

Q3

(1*10=10)

Write Algorithm to construct SLR parser. Show all the steps of SLR parsing for following grammar.

$S \rightarrow R$

$R \rightarrow Rb$

$R \rightarrow a$

Q4 Attempt both parts

(2*5=10)

- Draw the finite automata, which can be used by lexical analyzer to recognize identifier, number and operators.
- What is operator grammar? Write operator precedence parsing table for following grammar:

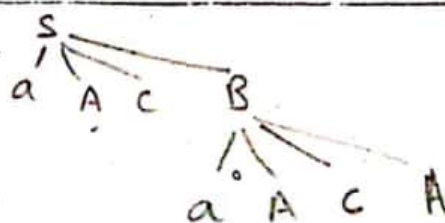
$E \rightarrow E+E | E * E | (E) | a$

$S \rightarrow aAcB | Bds$

$B \rightarrow aAcA | cAB | b$

$A \rightarrow aB | aBc | a$

aacbbcc



Ms. Shahu
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